

Salient Characteristics – Virtual Air Defense Systems Integrator Software

Salient Characteristics – V15 Virtual ADSI (or Equal)

The offered product shall meet or exceed the following minimum salient characteristics. These characteristics are derived from the documented capabilities of the Ultra Electronics V15 Virtual ADSI tactical data link system.

1. Virtualized Tactical Data Link Processing

- Must provide a **virtual machine–based implementation** of a tactical data link (TDL) processor.
- Must operate on a **VMware ESXi** hypervisor environment.

“The ADSI V15 virtual software is hosted on a VMware ESXi server...”

2. Dual-Virtual-Machine Architecture

- Must operate using **two virtual machines**:
 - One **Windows 7/10 Enterprise** VM for the C2/situational awareness display.
 - One **Red Hat Enterprise Linux** VM for data link processing.

“Two VMs host the ADSI V15 virtual software; a Windows 7/10 Enterprise VM and a Red Hat Enterprise Linux VM.”

3. Tactical Data Link Support

The system must support **both Ethernet and serial** tactical data link interfaces, including:

Link 16

- JREAP C (3011C)
- MTC
- MTDS
- SIMPLE

- MIDS on IP (LVT-1D, LVT-2, LVT-11, STT)

Link 11

- SIMPLE
- Indianhead
- Link 11B

Other Required Data Links

- VMF
- ATDL-1
- NATO Link-1
- MTCD
- Intel links including FDL, OTH-T, CMF, USMTF

“Core offering supports Link 16 and Link 11 Ethernet links...”

“Link 16 JREAP 3011C... Link 11 SIMPLE... Link 11B... Intel data links including FDL, OTH-T, CMF and USMTF.”

4. Compliance with Military Standards

- Must comply with **MIL-STD-6016** for Link 16 interoperability.

“With strict adherence to... military standards, including the latest MIL-STD-6016...”

5. Optional Serial-to-IP and ATDS/NTDS Bridge Integration

The system must support **either or both** of the following:

SyncNet8® Serial-to-IP Appliance

- Must support **up to eight serial TDL connections** over a single Ethernet interface.

“Supports up to eight serial data link connections to the SyncNet8 appliance...”

PowerNet™ ATDS/NTDS Bridge

- Must support **up to two Link 11 connections**.
- Must provide **redundant dual-port** operation.

- Must mitigate **cable length limitations** inherent in ATDS/NTDS.

“Supports up to two data link connections to the PowerNet appliance... designed with redundant capability... solves the problem of cable length limitations.”

6. Full ADSI Feature Set

The system must include:

- Dynamic Link Reconfiguration
- Drag-and-drop Configuration Wizard
- Windows-based System Manager
- C2 and situational awareness display capability

“All of the ADSI functionality is available... including Dynamic Link Reconfiguration, the drag-and-drop Configuration Wizard and Windows-based System Manager.”

7. Information Assurance

- Must include **robust information assurance** features suitable for secure tactical data link operations.

“Robust Information Assurance functionality.”

8. Minimum Hardware Compatibility Requirements

The system must be compatible with:

- 64-bit x86 processors
- Minimum **two CPU cores**
- NX/XD bit enabled
- **8 GB RAM per VM**
- One or more **1 GigE or 10 GigE** network interfaces

“Host machine with at least two core processors... 8 GB RAM per VM... One or more 1 GigE or 10 GigE Ethernet controllers.”